# StegoCNNv5: A Dual-Branch CNN for Multiclass Steganography Algorithm Detection (June 2026)

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*Abstract*— **The widespread adoption of digital steganography techniques poses a significant challenge to cybersecurity and digital forensics, as hidden communication channels are increasingly exploited for malicious purposes. Existing steganalysis approaches are predominantly binary classifiers capable of distinguishing only between cover and stego images, leaving the identification of the specific embedding algorithm as an open problem. This paper presents StegoCNNv5, a dual-branch Convolutional Neural Network designed for five-class steganography algorithm detection. The proposed architecture processes two complementary feature representations simultaneously: spatial noise residuals computed via Spatial Rich Model (SRM) filters across all three color channels, and a block-wise Discrete Cosine Transform (DCT) feature map derived from image luminance. Both branches are fused through a Squeeze-and-Excitation (SE) attention module that adaptively weights the contribution of each representation. The model is trained and evaluated on a balanced dataset of 150,000 test images spanning five classes: Cover, LSB, DCT, PVD, and BPCS. StegoCNNv5 achieves an overall test accuracy of 90.21% and a macro F1-score of 0.9019, with near-perfect detection of DCT-embedded images (F1 = 0.9996) and strong performance across all steganographic classes.**

**INDEX TERMS - Deep learning, steganalysis of digital images, SRM, DCT, feature fusion, Encode, Decode, LSB, DCT, PVD, BPCS.**

## I. INTRODUCTION

Steganography is the practice of concealing information within a digital medium in a manner imperceptible to an unintended observer [1]. Unlike cryptography, which renders a message unreadable, steganography hides the very existence of a communication channel, making it a particularly dangerous tool when exploited for covert operations, malware command-and-control, or illegal data exfiltration [2]. Among the most widely studied spatial-domain techniques are Least Significant Bit (LSB) substitution [3], Pixel Value Differencing (PVD) [4], and Bit-Plane Complexity Segmentation (BPCS) [5], while frequency-domain methods such as DCT-based embedding represent a distinct family of algorithms operating in the transform space [6].

Traditional steganalysis relies on handcrafted statistical features, with the Spatial Rich Model (SRM) [7] being one of the most influential frameworks, producing high-dimensional feature vectors that capture pixel prediction errors across multiple filter responses. However, the manual design of such features is labor-intensive and may fail to generalize across diverse embedding algorithms. The advent of deep learning has shifted the field toward end-to-end learnable detectors: XuNet [8] and SRNet [9] demonstrated that CNNs could match or surpass SRM-based classifiers for binary steganalysis. Nevertheless, most existing deep learning approaches remain binary classifiers, determining only whether an image has been modified without identifying the algorithm responsible.

This paper addresses that gap by proposing StegoCNNv5, a multiclass CNN steganalyzer capable of simultaneously detecting the presence of steganographic content and identifying which of four embedding algorithms was used. The model adopts a dual-branch architecture that processes SRM residuals and DCT-domain features in parallel, fusing them through a Squeeze-and-Excitation attention mechanism. This design exploits the complementary nature of spatial and frequency representations: SRM residuals capture localized pixel-level anomalies characteristic of spatial-domain methods such as LSB and PVD, while DCT features expose spectral distortions introduced by frequency-domain embedding.

The remainder of this paper is organized as follows. Section II describes the dataset generation pipeline, the feature extraction strategy, the StegoCNNv5 architecture, and the training configuration. Section III presents the quantitative evaluation on the held-out test set. Section IV discusses the results and their implications. Section V concludes the paper and outlines directions for future work.

## II. PROPOSED MODEL

### A. Data Generation

The dataset was constructed from cover images resized to 256×256 pixels using nearest-neighbor interpolation (`Image.NEAREST`), a deliberate choice that preserves the original pixel integer values without introducing blending artifacts that could corrupt the subtle steganographic signals embedded in subsequent steps. For each cover image, four stego variants were generated by applying LSB, DCT, PVD, and BPCS embedding independently, with each algorithm filling its maximum available payload using a mixed-vocabulary random text generator to ensure content variability across samples.

All algorithms operate exclusively on the blue channel of the BGR color space. LSB encodes one bit per pixel through direct substitution of the least significant bit; DCT encodes one bit per 8×8 block by comparing two mid-frequency coefficients at positions (4,3) and (3,4); PVD encodes between one and six bits per adjacent pixel pair according to a six-range difference table; and BPCS encodes data in bit-planes 2 and 3 of the blue channel using Gray code and block conjugation to ensure minimum complexity. Table I summarizes the key properties of each algorithm.

The complete dataset comprises **775,000 images** distributed across three phases. The training set contains 500,000 images (100,000 per class), the validation set contains 125,000 images (25,000 per class), and the test set contains 150,000 images (30,000 per class). All five classes — Cover, LSB, DCT, PVD, and BPCS — are perfectly balanced within each phase, eliminating class imbalance as a confounding factor during both training and evaluation.

| Algorithm | Domain | Channel | Embedding Unit | Capacty |
|---|---|---|---|---|
| LSB | Spatial | Blue | 1 bit/pixel | ~65,504 |
| DCT | Frequency | Blue | 1 bit/8x8 block | ~1,023 |

| | | | | |
|---|---|---|---|---|
| PVD | Spatial | Blue | 1-6 bits/pair | Variable |
| BPCS | Bit-plane | Blue | 64 bits/block | ~(Mx64-96)/65 bits |

B. Feature Extraction (Pre-processing Branch Inputs)

Rather than feeding raw pixel values to the network, each input image is transformed into two complementary feature representations before entering the dual-branch architecture, following the established practice of preprocessing-assisted steganalysis [7].

SRM Residuals. Five high-pass prediction-error kernels derived from the Spatial Rich Model [7] are applied independently to each of the three RGB channels, producing 15 residual maps per image. Each residual is computed via convolution with reflect-mode padding and subsequently truncated to the range [−T, T] with T = 4.0 and normalized to [−1, 1]. This preprocessing suppresses image content while amplifying the low-amplitude noise-like traces left by embedding operations, making the residuals highly sensitive to pixel-level manipulation.

DCT Feature Map. The input image is converted to a luminance channel Y using the standard ITU-R BT.601 coefficients (0.299R + 0.587G + 0.114B), mean-centered by subtracting 128. A 2-D DCT with orthonormal normalization is then applied block-wise on non-overlapping 8×8 regions. The resulting coefficients are mapped through a $\log(1 + |\cdot|)$ compression and globally normalized to [0, 1], yielding a single-channel feature map that encodes the energy distribution across spatial frequencies. This representation is particularly sensitive to frequency-domain embedding schemes such as the DCT steganography algorithm used in this work.

C. Architecture **—StegoCNNv5**

StegoCNNv5 is a dual-branch CNN that processes the SRM residual tensor (15 × 256 × 256) and the DCT feature map (1 × 256 × 256) along independent pathways before merging them through an attention-weighted fusion module.

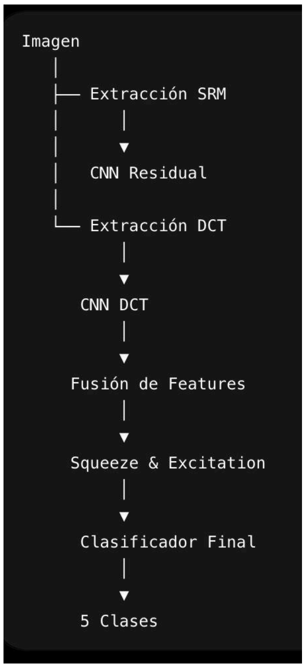


Fig. 1. CNN's simple architecture

**SRM BRANCH:**

The SRM branch begins with a stem block ConvBNMish(15→32) — a convolution followed by Batch Normalization and the Mish activation function — which projects the 15-channel residual input into a 32-channel feature space. The branch then passes through five successive Residual Blocks with interleaved Average Pooling layers (stride 2), progressively increasing channel depth: 32→32→64→96→128→128. Each Residual Block follows the standard skip-connection pattern with a learned shortcut projection when channel dimensions change. Global Average Pooling at the end of the branch collapses the spatial dimensions, producing a 128-dimensional feature vector.

**DCT BRANCH:**
The DCT branch begins with a lightweight stem Conv(1→16) → BatchNorm → Mish that expands the single-channel DCT map to 16 feature maps. Four Depthwise Separable Convolution blocks (SeparableConv2d) with interleaved Average Pooling layers progressively increase the channel depth: 16→32→64→96→128→128. Depthwise separable convolutions reduce parameter count and computational cost relative to standard convolutions while maintaining representational capacity. Global Average Pooling produces a 128-dimensional feature vector analogous to the SRM branch output.

**FUSION + SE:**
The two 128-dimensional branch outputs are concatenated along the channel dimension to form a 256-dimensional joint representation. This vector is reshaped into a 256×1×1 tensor and passed through a Squeeze-and-Excitation (SE) module [10] with reduction ratio r = 8. The SE module performs global average pooling (squeeze), followed by two fully connected layers with ReLU and Sigmoid activations (excitation), producing a 256-dimensional channel attention vector. This vector re-scales the fused representation, allowing the network to adaptively emphasize features from whichever branch — SRM or DCT — is more discriminative for the class being predicted.

**CLASSIFICATION HEAD:**
The attention-scaled 256-dimensional vector is passed through a three-layer MLP classification head: Linear(256→128) → BatchNorm → Mish → Dropout(0.45) → Linear(128→64) → BatchNorm → Mish → Dropout(0.45) → Linear(64→5). The output layer produces five unnormalized logits corresponding to the classes Cover, LSB, DCT, PVD, and BPCS. A dropout rate of 0.45 is applied before each hidden linear layer to mitigate overfitting.

D. Training Configuration

The model was optimized using AdamW [11] with an initial learning rate of $3\times10^{-3}$ and a weight decay of $1\times10^{-4}$. The learning rate was annealed over 20 epochs using a Cosine Annealing schedule with a minimum learning rate of $3\times10^{-5}$. An explicit L2 regularization term with coefficient $\lambda = 1\times10^{-4}$ was added to the loss at each training step. The loss function is weighted Cross-Entropy, where per-class weights were computed using the balanced strategy from sklearn.utils.class_weight.compute_class_weight to account for any residual class imbalance. Training was performed with a batch size of 64 on a CUDA-enabled GPU. The model checkpoint with the highest macro F1-score on the validation set was selected for final evaluation.

| **Hyperparameter** | **Value** |
|---|---|
| Optimizer | AdamW |
| Initial Learning Rate | 3e-5 |
| LR Schedule | Cosine Annealing |
| Minimum LR | 3e-5 |
| Weight Decay | 1e-4 |
| L2 Regularization (lambda) | 1e-4 |
| Dropout | 0.45 |
| Batch Size | 64 |
| Epochs | 20 |
| Loss Funtion | Weighted CrossEntropyLoss |
| Model Selection Criterion | Best Macro F1 (Validation) |

## III. EVALUATION

[texto]

| | Precision | Recall | F1-Score |
|---|---|---|---|
| Cover | 0.7998 | 0.8168 | 0.8082 |
| LSN | 0.8531 | 0.8099 | 0.8309 |
| DCT | 0.9996 | 0.9995 | 0.9996 |
| PVD | 0.9598 | 0.9584 | 0.9591 |
| BPCS | 0.8979 | 0.9256 | 0.9115 |
| Accuracy | | 0.9021 | |
| Macro F1-Score | | 0.9019 | |

Table. 1. Evaluation results

# IV. DISCUSSION

## A. Per-Class Performance Analysis

The substantial performance gap between classes reveals that detectability is directly related to the nature and magnitude of the modifications introduced by each algorithm.

DCT steganography achieves near-perfect classification (F1 = 0.9996) because its embedding mechanism introduces structured artifacts in mid-frequency DCT coefficients that are highly consistent and spatially regular. Since the DCT branch of StegoCNNv5 is specifically designed to capture frequency-domain anomalies through block-wise log-compressed coefficient maps, it effectively acts as a matched filter for this class of distortion.

PVD (F1 = 0.9591) and BPCS (F1 = 0.9115) are also detected with high reliability. PVD modifies pixel pairs in a content-adaptive manner — embedding more bits in high-gradient regions — which produces statistically detectable asymmetries in pixel difference distributions. BPCS operates on bit-planes 2 and 3 of the blue channel, replacing complex regions with pseudo-random data; while conjugation ensures local complexity is maintained, the alteration of the binary structure across multiple bit-planes leaves traces that the SRM branch captures effectively.

LSB substitution (F1 = 0.8309) is the most challenging stego class to detect, as it introduces the smallest possible perturbation — a single bit per pixel — producing noise-like residuals with no spatial structure. The Cover class (F1 = 0.8082) presents the lowest F1-score not because unmodified images are inherently difficult to classify, but because a fraction of Cover images are misclassified as LSB, suggesting that some natural images contain pixel patterns statistically similar to LSB-embedded content.

The relationship between embedding capacity and detectability can be described in terms of the signal-to-noise ratio of the introduced distortion. A higher payload relative to image content results in greater statistical deviation from the cover distribution, which can be formalized as:

$$D_{KL}(P_{stego} \| P_{cover}) \propto \frac{b}{N}$$

Ec. 1.

where DKLD_{KL}DKL denotes the Kullback-Leibler divergence between the stego and cover pixel distributions, bb

b is the number of bits embedded, and NN

N is the total number of available embedding units. This relationship is consistent with the observed ordering of per-class F1-scores: DCT and BPCS, which produce larger or more structured distortions, are detected more reliably than LSB, which operates at the minimum possible modification magnitude.

## B. Limitations

Several limitations constrain the generalizability of the current system. First, all four steganographic algorithms embed data exclusively in the blue channel, which simplifies detection but does not reflect real-world implementations that may operate across all channels or use adaptive channel selection. Second, the training pipeline always fills the maximum available payload capacity; in practice, partial payloads produce weaker statistical traces and are expected to degrade detection performance. Third, the dataset consists entirely of lossless-format images (PNG, BMP, TIF); JPEG compression, which modifies DCT coefficients during encoding, would significantly alter the feature distributions the model has learned and represents an important domain shift not addressed in this work.

## V. CONCLUSION

This paper presented StegoCNNv5, a dual-branch Convolutional Neural Network for multiclass steganography algorithm detection. Unlike conventional binary steganalyzers, the proposed model simultaneously determines whether an image has been modified and identifies which of four embedding algorithms — LSB, DCT, PVD, or BPCS — was responsible. The architecture exploits two complementary feature representations: SRM residuals that amplify spatial pixel-level anomalies, and block-wise DCT feature maps that expose frequency-domain distortions, fused through a Squeeze-and-Excitation attention module that adaptively weights each branch's contribution.

Evaluated on a balanced test set of 150,000 images, StegoCNNv5 achieves an overall accuracy of 90.21% and a macro F1-score of 0.9019, with near-perfect detection of DCT-embedded images (F1 = 0.9996) and strong performance across PVD and BPCS classes. The results confirm that detectability is strongly correlated with the magnitude and structure of the embedding-induced distortion, with LSB posing the greatest challenge due to its minimal single-bit perturbation.

Future work will focus on three directions: (1) extending the model to partial-payload scenarios, where embedding strength is reduced below maximum capacity; (2) evaluating generalization to JPEG-compressed images, where DCT quantization introduces additional noise that may mask or alter steganographic traces; and (3) incorporating adaptive channel selection to handle implementations that embed across all three color channels rather than exclusively in the blue channel.

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